

Pedestrian Intention and BanglaTesla Trajectory Prediction for Unstructured Traffic Environments in Bangladesh

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Pedestrian Intention Prediction (PIP)

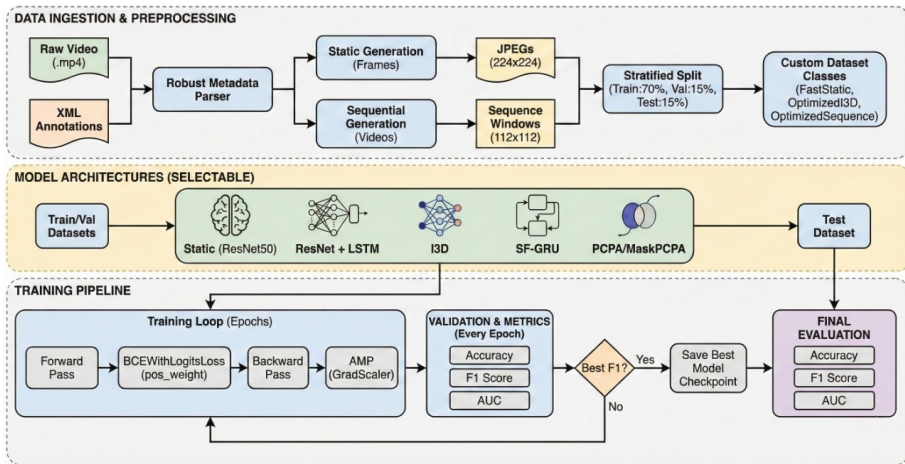


Fig 1. PIP Baseline Pipeline

PIP Baseline Comparison

Model	JAAD		PIE		IDD-PeD	
	AUC	F_1	AUC	F_1	AUC	F_1
Static	0.75	0.55	0.60	0.41	0.8736	0.8335
ConvLSTM	0.57	0.32	0.55	0.39	0.7560	0.7669
C3D	0.81	0.65	0.67	0.52	0.7624	0.7776
I3D	0.74	0.63	0.73	0.62	0.7609	0.7716
SF-GRU	0.77	0.53	0.79	0.69	0.7003	0.7290
SingleRNN	0.77	0.54	0.77	0.67	0.7209	0.7674
PCPA	0.79	0.56	0.86	0.77	0.6904	0.7216
MaskPCPA	0.82	0.63	0.86	0.80	0.6886	0.7278

Table: II. Comparison of AUC and F1 scores across JAAD, PIE, and IDD-PeD datasets.

Pedestrian Trajectory Prediction Pipeline

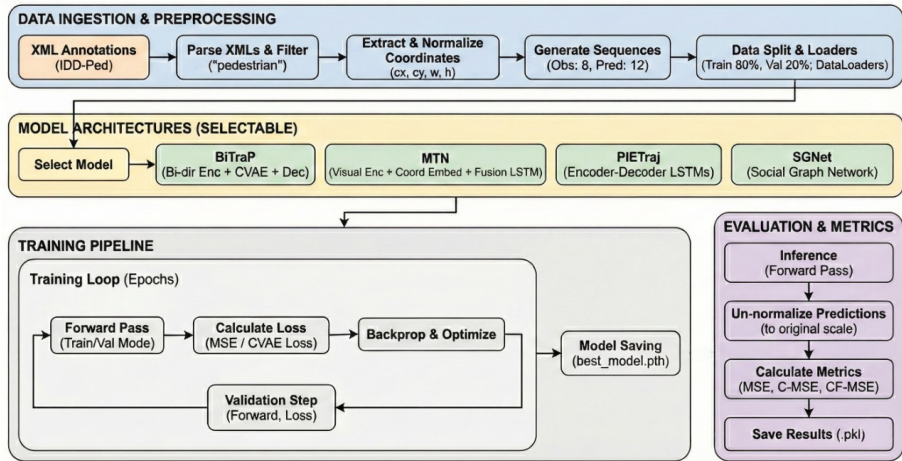


Fig 2. PTP Baseline Pipeline

PTP Baseline Comparison

Method	MSE	C-MSE	CF-MSE
PIETraj	5269.31	20297.43	19779.86
MTN	5477.66	16728.28	33456.56
BiTraP	6191.42	19779.86	39559.71
SGNet	13557.26	42422.72	84845.43

Table: III: Evaluation of PTP baselines on IDD-PED datasets

PIP on Scenario Pipeline

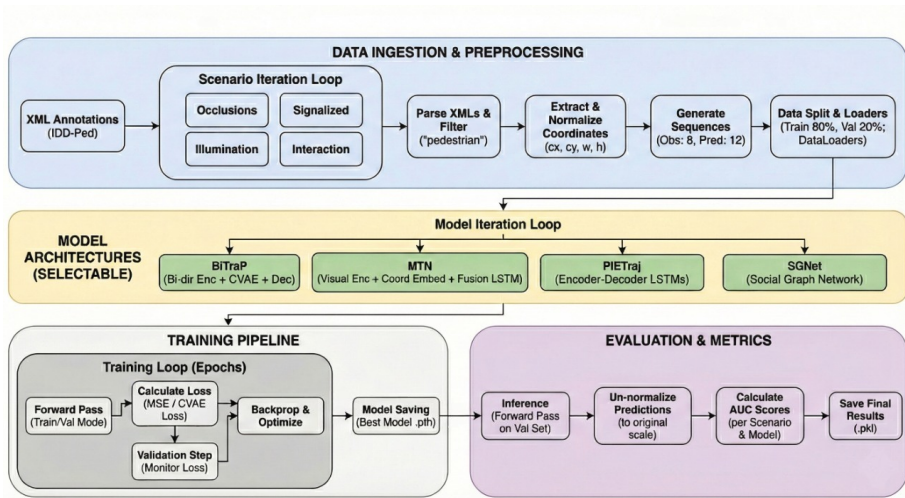


Fig 3. PIP Scenario Baseline Pipeline

PIP baselines on Occlusions, Signalized, Illumination, and Interactions

Method	Occlusion		Signalized		Illumination		Interaction	
	w/o	w	Yes	No	Day	Night	w/o	w
SingleRNN	0.52	0.5	0.52	0.54	0.5	0.58	0.52	0.54
I3D	0.92	0.5	0.92	0.94	0.92	0.92	0.8	0.88
SF-GRU	0.46	0.5	0.45	0.54	0.42	0.47	0.47	0.53
PCPA	0.5	0.5	0.5	0.5	0.51	0.52	0.5	0.49

Table: Evaluation of PIP baselines on Occlusions, Signalized Types, Illumination Changes, and Vehicle-Pedestrian Interactions in IDD-PED dataset.

PTP Scenario Pipeline

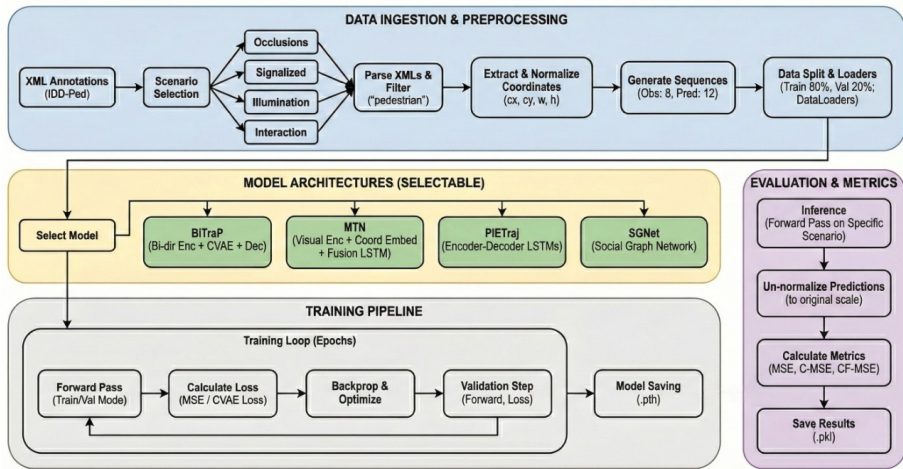


Fig 4. PTP Scenario Pipeline

PTP baselines on Occlusions, Signalized, Illumination, and Interaction

Method	Occlusion		Signalized		Illumination		Interaction	
	w/o	w	Yes	No	Day	Night	w/o	w
PIETraj	4630	5807	5270	4522	5274	4947	5135	5382
MTN	5208	6003	5545	5777	5946	4739	5589	5500
BiTraP	5480	8391	6876	6475	7630	5007	6823	6830
SGNet	4850	6380	5547	5623	6046	4454	5563	5520

Table: Evaluation of PTP baselines on Occlusions, Signalized Types, Illumination Changes, and Vehicle-Pedestrian Interactions in IDD-PED dataset.

BanglaTesla Trajectory Prediction Pipeline

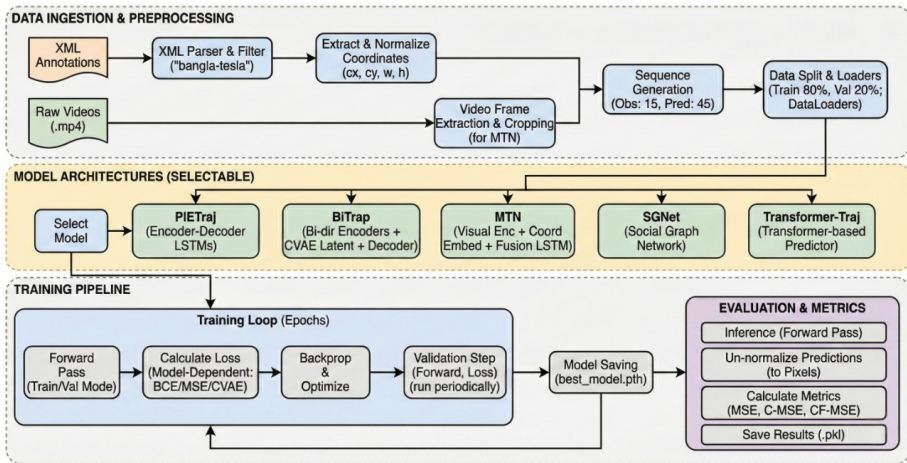


Fig 5. BTP Baseline Pipeline

BanglaTesla Trajectory Prediction Baseline Comparison

Method	MSE	C-MSE	CF-MSE
MTN	23359.17	59450.51	128929.07
BiTraP	20849.88	50561.20	112180.95
PIETraj	19385.38	51930.15	113628.50
SGNet	13872.15	28258.27	63870.88
Transformer-Traj	43679.06	109290.25	221782.73

Table: Evaluation of trajectory prediction models using MSE, C-MSE, and CF-MSE.